MIL-G-96A
1 August 1969
SUPERSEDING
JAN-G-96
25 October 1944

MILITARY SPECIFICATION

GUM TRAGACANTH
(For Use in Ammunition)

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers one type of gum tragacanth for use in ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D1193-66 - Specification for Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

3. REQUIREMENTS

3.1 Material. Gum tragacanth shall be in the form of clear fine ribbons or white flakes.

3.2 Chemical requirements. The material shall conform to requirements indicated in table I.

Table I. Chemical properties

Properties	Requirements	Test para.
Chlorine or chlorides	none	4.4.2
Acidity, percent	0.40 max.	4.4.3
Foreign gums, percent	none	4.4.4
Indian gums, percent	none	4.4.5
Ash, percent	3.5 max.	4.4.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Size of lot.

4.2.1 At place of manufacture. For the purpose of sampling, a lot of gum tragacanth shall consist of a manufacturer's batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform. If the material cannot be identified by batch, a lot shall consist of not more than 1,000 pounds of gum tragacanth offered for delivery at one time.

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4.2.2 At place of delivery. For the purpose of sampling, a lot of gum tragacanth shall consist of all the gum in a single shipment.

4.3 Sampling. For chemical tests, a 1/4 pound representative specimen shall be removed from each of five containers taken at random from each lot. The specimen shall be placed in a clean, dry container, thoroughly blended, and labeled to identify the container and lot represented.

4.4 Test methods.

4.4.1 General. Distilled water in accordance with ASTM D1193 and analytical grade reagents shall be used throughout the tests. Blank determinations shall be run and corrections shall be applied when necessary.

4.4.2 Chlorine or chlorides. Shake about 2 gm. of the sample with 100 ml. of distilled water until the gum is fully swollen and free from lumps. Add 100 ml. of distilled water and agitate the solution thoroughly. To 100 ml. of this solution, add about 10 ml. of approximately N/10 silver nitrate solution. No opalescence or white cloudiness should result. To the remaining 100 ml. of the gum solution, add approximately 5 ml. of 10 percent potassium iodide solution, freshly prepared. Agitate thoroughly and add 2-3 ml. starch indicator solution. No blue coloration should result.

4.4.3 Acidity. Treat approximately 1 gm. of the sample with 100 ml. of distilled water until the gum is evenly dispersed. Add 0.1 ml. of a 1 percent phenolphthalein solution and titrate with N/10 sodium hydroxide solution. Calculate the acidity as acetic acid contents as follows:

$$\text{Percent acetic acid} = \frac{6.0\ VN}{W}$$

where: V = ml. of sodium hydroxide solution used.
N = normality of the sodium hydroxide solution.
W = weight of sample.

4.4.4 Foreign gums. Weigh out about 2 gm. of the sample. Shake with 100 ml. of distilled water until it is thoroughly swollen and free from lumps. Add to the mixture approximately 2 gm. of sodium tetraborate ($Na_2B_4O_7 \cdot 10\ H_2O$). Shake the mixture thoroughly until the salt is completely dissolved. No loss in transparency and no change in consistency should be evident after standing for 24 hours.

4.4.5 Indian gums. Form a mucilage with approximately 1 gm. of the sample and 20 ml. distilled water. Add 5 ml. of hydrochloric acid (sp. gr. 1.2). Boil the mixture for 5 minutes. No pink or red color shall appear.

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4.4.6 Ash. Weigh approximately 10 gm. of the sample in a tared porcelain or silica crucible. Heat over a free flame for about 30 minutes. Cool in a desiccator and weigh. Heat for an additional time, to be sure that constant weight has been reached. Cool and weigh. Calculate the gain in weight of the crucible as percentage of ash.

4.5 Rejection. A lot shall be rejected for failure to comply with the requirements of section 3.

5. PREPARATION FOR DELIVERY

5.1 Packing. Unless otherwise specified (see 6.2), 300 pounds plus or minus 2 pounds of gum tragacanth shall be packed in a fiber drum which complies with requirements of the Uniform Freight Classification, or with National Motor Freight Classification ratings, and regulations applicable to the mode of transportation. Containers shall be capable of being stacked and of supporting superimposed loads during shipment and shall be adequate to assure safe delivery to destination without damage to contents and acceptable for transport at the most favorable rate.

5.2 Marking. In addition to any special marking required by the contract or order, marking for identification and shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Gum tragacanth is intended for use as an ingredient of primer compositions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Packing (see 5.1)

Custodian:
Army - MU
Navy - OS

Review activities:
Army - MU, MD
Navy - OS

User activities:
Army - MI, MU

Preparing activity:
Army - MU

Project No. 6810-0581

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